

Gunstuff Ballistics App -- User Manual

1. Overview

Gunstuff Ballistics is a mobile application for managing rifles, venues, shooting sessions, and load development (Ladder & OCW), with tools for wind drift calculation and full backup/restore support.

This manual explains the normal day■to■day workflow of the app.

2. Main Menu

From the main menu you can access:

- Start Session\
 - Session Summaries\
 - Load Development\
 - Tools & Utilities
-

3. Rifles & Load Data

3.1 Adding a Rifle

Open ****Rifles**** → Tap ****Add****

Enter: - Name - Caliber - Barrel length + unit - Twist rate - Muzzle velocity - Scope + unit - Round count - Notes (optional) - Rifle photo (optional)

Tap ****Add rifle**** or ****Save changes****.

3.2 Editing / Deleting a Rifle

Select the rifle from the dropdown → Tap ****Edit**** or ****Delete****.

3.3 Rifle Photos

You may capture a rifle photo. Photos are stored on the device filesystem to reduce storage usage.

3.4 Load Data (per rifle)

Each rifle can have multiple loads.

Tap **Show load data** → **Add**

Enter: - Powder - Charge (grains) - Lands (Ogive) - COAL + COAL unit (mm / in) - COAL (Ogive) - Primer - Bullet + weight - BC - Notes

The **Jump** value is calculated automatically.

Tap **Add load** or **Save load**.

> The default COAL unit is taken from Preferences when a new load is
> created.

3.5 Editing / Deleting Loads

Use the **Edit** and **Delete** buttons next to each load.

4. Venues & Subranges

4.1 Adding a Venue

Open **Venues** → **Add**

Fields: - Name - Location - Altitude - Notes

4.2 Subranges

You may add multiple subranges per venue:

Each subrange has: - Name - One or more distances (comma-separated)

Example: 500, 600, 780

4.3 Editing / Deleting Venues

Select venue → ****Edit**** or ****Delete****

5. Sessions (Shooting Wizard)

Sessions are recorded using a step■by■step wizard.

Step 1 -- Setup

Required: - Session title - Rifle - Venue

Optional: - Subrange

Step 2 -- Environment

Enter manually or use Kestrel sync: - Temperature - Pressure - Humidity - Wind speed - Wind direction

Step 3 -- Shots

Enter: - Distance - Planned shots - Shot data - Notes

Step 4 -- Complete

The session becomes read■only and moves to History.

If required fields are missing, the app will prevent completion.

6. Load Development

6.1 Creating a Project

Open ****Load Development**** → Select rifle → ****New****

Choose: - Project name - Type: Ladder or OCW - Powder - Bullet - Weight - Lands / COAL - Start / End / Step charge

Tap ****Save & plan charges****.

6.2 Recording Results

Enter velocities per charge / group.

Graphs are generated automatically.

6.3 Results & Graphs

You may toggle: - Graph view - Results table

6.4 Editing

You may edit velocities and notes at any time.

7. History

The History screen shows:

- Pending sessions
- Completed sessions
- Search by venue

You can: - Open sessions - Continue unfinished sessions - Delete sessions

8. Wind Effect Tool

This tool calculates wind drift.

Inputs: - Rifle - Range - Muzzle velocity - Wind speed - Wind direction

Outputs: - Drift in MIL / MOA - Drift in cm / inches

Includes a clock■style wind direction dial.

9. Preferences

Preferences control:

- Distance units
- Wind units
- Load development COAL unit (mm / in)
- Other UI defaults

Preferences are stored locally on the device.

> Preferences are NOT included in JSON backups.

10. Backup & Restore

Export

Tools → Export / Import → Export

Creates a JSON file containing: - Rifles - Loads - Venues - Sessions - Load Development projects

Import (Merge Mode)

Import uses ****merge rules****:

- Existing rifles (matched by name) are NOT overwritten
- New rifles are added
- Sessions are added if not duplicates
- Load development projects add missing entries only
- Preferences are NOT imported
- Rifle filesystem photos are NOT restored

11. Known limitations

- Preferences are device■local
 - Filesystem rifle photos cannot be restored on another device
 - Import never overwrites existing data
-

12. Recommended workflow

1. Create rifles
2. Create venues + subranges
3. Capture sessions
4. Use Load Development for tuning
5. Backup regularly
6. Export rifle data PDFs when needed

13. Support & troubleshooting

If data appears missing after import: - Check if the rifle already existed (merge rule) - Check Preferences - Check rifle photos (filesystem)

End of manual.